

DESIGN CRITERIA			SIZING (TAS DESIGN LOADS)		
	Heating	Cooling		Heating	Cooling
Room set point	none	none	Design load	0 Btu/h	0 Btu/h
Outdoor dry bulb	—	—	Design load per area	0.00 Btu/h·ft ²	0.00 Btu/h·ft ²
Outdoor RH (coincident)	—	—	Sizing multiplier	not set	not set
ROOM HUMIDITY			Design loads: from Tas sizing — date/currency not recorded		
Humidification set point lower RH limit	—				
Dehumidification set point upper RH limit	—				

HEATING

Design day	Leeds_TRY ANN HTG 100% CONDS DB	
	Design day	Full year
Peak sensible load	7,738 Btu/h	2,737 Btu/h
Peak hour	15:00–16:00	1 Jan 00:00–01:00 (HOY 1)
Room dry bulb	59.1 °F	56.1 °F
Room resultant	55.8 °F	54.7 °F
Room RH	21 %	84 %
Room humidity ratio	2.2 g/kg	8.1 g/kg
Outdoor dry bulb	—	37.9 °F
Outdoor RH	—	100 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (Btu/h)	Full year (Btu/h)
Occupancy (sensible)	0	358
Equipment (sensible)	0	290
Infiltration / ventilation	-1,096	-526
Building heat transfer	-2,266	-1,114
External conduction — opaque	-3,053	-1,118
External conduction — glazing	-1,323	-627

Zero at both peaks (not listed): Solar, Lighting, Air movement (inter-zone), Air handling unit

LATENT GAINS AT THE SENSIBLE PEAK HOUR

Component	Design day (Btu/h)	Full year (Btu/h)
Occupancy (latent)	0	263

Zero at both peaks (not listed): Equipment (latent)

Latent gains: internal gains only (occupancy, equipment) at the hour of the sensible peak. The room's latent load, which also includes moisture from infiltration and ventilation, is not in the results and is not derived.

Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

COOLING

Design day	Leeds_TRY ANN CLG 0% CONDS DB=>GRad	
	Design day	Full year
Peak sensible load	6,734 Btu/h	6,729 Btu/h
Peak hour	00:00–01:00	3 Jul 19:00–20:00 (HOY 4412)
Room dry bulb	67.6 °F	66.1 °F
Room resultant	70.6 °F	69.1 °F
Room RH	100 %	58 %
Room humidity ratio	14.5 g/kg	8.0 g/kg
Outdoor dry bulb	—	63.5 °F
Outdoor RH	—	59 %

SENSIBLE LOAD COMPONENTS AT PEAK

Component	Design day (Btu/h)	Full year (Btu/h)
Solar	0	2,852
Lighting	0	512
Occupancy (sensible)	358	512
Equipment (sensible)	290	1,535
Infiltration / ventilation	-29	-180
Building heat transfer	3,315	1,017
External conduction — opaque	3,064	748
External conduction — glazing	-264	-267

Zero at both peaks (not listed): Air movement (inter-zone), Air handling unit

LATENT GAINS AT THE SENSIBLE PEAK HOUR

Component	Design day (Btu/h)	Full year (Btu/h)
Occupancy (latent)	263	375

Zero at both peaks (not listed): Equipment (latent)

Latent gains: internal gains only (occupancy, equipment) at the hour of the sensible peak. The room's latent load, which also includes moisture from infiltration and ventilation, is not in the results and is not derived.

Components as the simulation reports them at each peak hour: + gain to the room air, – loss from it. No total is derived.

RESULTS

	Heating	Cooling
Status	Available	Available
Result source	SAM.Analytical.Tas	SAM.Analytical.Tas
Read into model	27 Sep 2026 20:42	27 Sep 2026 20:42
Matches current model	not recorded	not recorded

Peak hour: a design-day peak has an hour of the day only, no date; a full-year peak shows day and hour, no year, and its hour of the year (HOY 1 = 1 Jan 00:00–01:00, HOY 8760 = 31 Dec 23:00–24:00).

Design loads (sizing) and simulated peaks are shown side by side for information; no acceptance rule is applied.